

Cover letter

To: The Editors, *The Computer Journal*

Re: Submission of *Protocol-Governed Computing: Business Software That Runs on a Domain-Neutral Substrate*

Dear Editors,

I submit the manuscript above for consideration as a regular paper in *The Computer Journal*. It is original, is not under consideration elsewhere, and has not been published previously. I am the sole author, Independent Researcher, unaffiliated with any organization. I have over forty years of experience in large-scale, business-critical systems integration and software architecture.

What the paper is

Operating systems admit a new process, virtual machines new bytecode, and database engines a new query without being rebuilt for it. Governed business domains are usually admitted differently: a new domain arrives as a change to the executable composition that already carries the other domains. The paper asks whether that coupling can be removed — whether a governed business domain can arrive as an executable artifact that a shared substrate admits and executes without acquiring behavior specific to that domain.

It makes the question testable by separating two conditions that are ordinarily conflated: **C1**, admission without substrate modification, and **C2**, no domain-specific behavior in the substrate. A criterion distinguishes them — a substrate contributes domain-specific behavior when the admitted artifact does not determine a domain-visible outcome and the substrate resolves it.

The paper then reports an architectural case study. Four heterogeneous domains — decentralized identity, AI governance and licensing, library catalog management, and a computational conformance workload — are compiled into sealed snapshots and executed by one traversal engine. Measurements taken by read-only inspection of the deposited artifacts show that no admission introduced domain-specific behavior into the substrate, that all 22 domain-specific capability implementations are namespaced to the domains that declare them, that no domain vocabulary participates in an execution decision, and that an injected cross-domain reference is refused during construction before any artifact is produced.

The substrate did change during the recorded admissions, always at the shared contract by which it consumes a snapshot. Literal C1 is therefore not demonstrated; the separately named condition C1' — admission without domain-specific substrate modification — is demonstrated. The paper also identifies three boundaries for further engineering: per-admission construction spillover was not measured, capability bindings do not fix the content identity of the code they name, and snapshot identity includes renderer-dependent images and therefore does not reproduce across environments.

Why *The Computer Journal*

The paper's subject is the boundary between an executable artifact and the substrate that consumes it — the same relation the journal's readership already understands in operating systems, virtual machines and database engines. Its argument is made in those terms and asks whether the separation transfers one layer up. It is an architecture and systems paper with measurements, not a business-process or workflow-management contribution, and it belongs with work that treats execution substrates as objects of study in their own right. **Section C is the intended fit.**

The paper is also written to be checked rather than believed. The composition it measures is deposited under a DOI, the specification it claims conformance to is deposited separately, and every reported figure was taken by a read-only inspector that ships with the composition. A reviewer who doubts a number can obtain the artifact and take it again.

What is novel

1. **A two-condition test applied at the business-domain layer.** Substrate independence is usually asserted informally. C1 and C2, together with the determination criterion, give a form that can be tested and potentially falsified. The paper applies the test directly and reports both the demonstrated C1' result and the boundary of the stronger C1 condition.
2. **The distinction between substrate evolution and domain coupling.** C1' separates a substrate that changes because a domain requires new behavior from one that changes because a shared contract is being evolved. This distinction is what the admission record actually supports and provides a precise way to state the result without conflating substrate change with domain coupling.
3. **Determination separated from realization.** The sealed artifact carries what is determined; the capability implementations carry what is performed. Keeping these apart is what makes both the C2 claim and the provenance gap precise rather than rhetorical.
4. **A refusal experiment rather than another demonstration.** An undeclared cross-domain reference was injected deliberately; the compiler refused it during construction, before producing an artifact, and recorded the refusal as an artifact of the same kind used to evidence an admission.

Four companion papers by the same author are under review elsewhere. All rest on the same architecture and the same reference realization, and Section 1.3 of the manuscript states each paper's question and method so that overlap can be checked rather than assumed. No text, argument, data or evidence from any of them is reused here.

Thank you for considering the manuscript.

Yours sincerely,

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Encl: Declaration of generative AI use - an extract from manuscript

Declaration of generative AI use

During the preparation of this work the author used Claude/Opus (Anthropic) in the four supporting capacities listed below. After using this tool the author reviewed, revised and edited the content as needed, and takes full responsibility for the content of the publication.

The author is the sole author. The author conceived and wrote the research content — the architectural argument, the two conditions, the measurements, the claims made and the limits placed on them. The generative AI tool used is not an author, and could not be: a tool cannot hold the accountability that authorship entails. The tool generated no research data, performed no derivation and conducted no analysis of its own.

1. **Drafting assistance under the author’s direction.** The assistant drafted and restructured prose from the author’s outline, argument and decisions. The author reviewed, revised and accepted every passage. No text entered the manuscript without that review.
2. **Rendering Figures 1, 2, 3 and 5.** The author supplied each figure’s content: the decomposition, the partitions, the properties and the placement of each property at its point of origin. The assistant produced the drawings and refined their layout. These are conceptual schematics. They present no research data, no experimental results and no data visualization, and no data, result or figure presenting data was generated or altered by any AI tool.
Figure 4 is not an AI product: it is rendered from the compiled snapshot by the composition’s own inspector, as §5.6 of the manuscript states.
3. **Extracting material from the reference realization.** The observations reported in Section 6 were read from the repositories, build outputs, snapshot inspections and execution traces of the reference realization and transcribed into the manuscript. The implementation, not the manuscript, is the record.
4. **Proofreading and copy-editing.**

A separate disclosure belongs to the reference realization itself, because it bears on Section 6. Parts of that realization were constructed by AI workers performing governed activity under the author’s direction, within the architecture the paper describes. The author held scope, admission and promotion authority throughout; the workers held none. That arrangement is not what this paper measures, and the results in Section 6 do not depend on it. A companion study examines it directly.

The same declaration appears in the manuscript, following the Conclusion.